Public Understanding Survey

Emerging biomedical technologies and Alzheimer's disease treatments



Q01 What is your age?

Single choice: select one

- ○ Under 18
- ○ 18-30 years
- ○ 31-45 years
- ○ 46-60 years
- ○ 61-70 years
- ○ 71-80 years
- ○ 81 years or older

Q02 Have you received education in life sciences, medicine, pharmacy, or a related field?

Single choice: select one

- ○ Yes; currently studying or working in a related field
- ○ Previously received relevant education; not currently working in a related field
- ○ No relevant educational background
- ○ Not sure

Q03 Have you or someone close to you had experience with Alzheimer's disease or another cognitive disorder?

Single choice: select one

- ○ Direct experience, e.g. personal or family diagnosis, or caregiving
- ○ Some exposure or knowledge, but no long-term caregiving experience
- ○ No relevant experience
- ○ Prefer not to answer

Q04 Before completing this survey, which option best describes your familiarity with RNA editing?

Single choice: select one

- ○ Never heard of it
- ○ Heard the term, but do not know what it means
- ○ Know its general uses or basic concepts
- ○ Can briefly explain its basic principles and possible uses
- ○ Quite familiar with the relevant technologies

Q05 For each statement below, indicate whether it matches your current understanding.

Select one response per statement

1. RNA editing can change information in specific RNA molecules without directly rewriting genomic DNA.

○ Mostly matches my understanding ○ Mostly does not match my understanding ○ Not sure

2. RNA editing methods may differ in duration of action, safety, and technological maturity.

○ Mostly matches my understanding ○ Mostly does not match my understanding ○ Not sure

3. Understanding the mechanism alone is usually insufficient to judge whether an RNA-editing therapy is suitable for a patient.

○ Mostly matches my understanding ○ Mostly does not match my understanding ○ Not sure

4. If an RNA-editing technology improves relevant measures in animal studies, it can usually be considered effective in patients.

○ Mostly matches my understanding ○ Mostly does not match my understanding ○ Not sure

Q06 Suppose a study of a novel RNA-editing treatment for Alzheimer's disease reports:

Multiple choice

After treatment, a disease-related biomarker decreased by an average of 40%.

Based on this finding alone, which of the following interpretations do you consider most appropriate?

- [ ] The treatment has already demonstrated fairly clear clinical benefit
- [ ] The treatment may have had its intended biological effect, but symptoms and long-term outcomes still need assessment
- [ ] The treatment is worth using; suitability for an individual patient mainly depends on a physician's judgment
- [ ] The information mainly shows a change in a study measure and is not yet sufficient to judge clinical value
- [ ] Not sure

Q07 The researchers subsequently release the following information:

Single choice: select one

The treatment has undergone preliminary human studies. Relevant biological measures improved in some patients, but the sample remains small. Further follow-up is needed to assess cognitive improvement and long-term adverse effects.

If you saw these findings, which option would best describe your assessment of the technology at this stage?

- ○ There is some evidence of research value, but sufficient clinical benefit still needs further verification
- ○ Biological changes in humans provide preliminary proof of therapeutic value
- ○ The most important next step is to confirm further improvement in disease-related biological measures
- ○ Whether it is worth further use mainly depends on the patient's acceptance of new technologies
- ○ I need more information before forming a judgment

Q08 If you or a family member were considering an unfamiliar new treatment for Alzheimer's disease, which information would you prioritize before making a decision?

Multiple choice: select up to 4

- [ ] How much it actually improves or slows the condition compared with existing treatments
- [ ] Types and probabilities of serious adverse effects
- [ ] The size and stage of the human studies supporting the conclusions
- [ ] Which patients may benefit and which may not be suitable
- [ ] Whether the molecular mechanism of action is sufficiently clear
- [ ] How long the treatment effect lasts
- [ ] What alternatives are available if this treatment is not chosen
- [ ] Professional reputation of the institution or development team
- [ ] Personal experiences of patients who have received the treatment
- [ ] Treatment costs and long-term financial burden

Q09 Which way of presenting information would best help you understand the risks of a new treatment?

Single choice: select one

- ○ "Current studies show generally good safety, but some risks remain."
- ○ "About 3 in every 100 treated participants experienced a specified serious adverse effect," together with the total study size and follow-up duration
- ○ "Serious adverse effects are uncommon, and most participants can tolerate the treatment."
- ○ "No safety issues beyond those anticipated by the researchers have been found so far."
- ○ I find it easier to judge when these types of information are combined
- ○ Not sure

Q10 When learning about an unfamiliar biomedical technology, which issues most hinder your understanding or judgment?

Multiple choice: select up to 3

- ☐ Too much technical jargon
- ☐ The stage of research is not stated
- ☐ It is called "effective" without specifying what improved
- ☐ Risk is described as "low" without numbers or probabilities
- ☐ Information is too long or lacks clear key points
- ☐ Charts or data are hard to understand
- ☐ It is unclear whether the information source is reliable
- ☐ No comparison with existing treatments or with no treatment
- ☐ No captions, audio, illustrations, or easy-read version
- ☐ No trusted person is available for further questions
- ☐ Other: ____
- ☐ No obvious barriers at present

Q11 If we create educational or informed-consent materials on this topic, which formats would help you understand them?

Multiple choice

- ☐ Concise one-page summary
- ☐ Illustrations or flowcharts
- ☐ Short videos
- ☐ Detailed explanations from experts
- ☐ Real numbers and examples of absolute risk
- ☐ Frequently asked questions
- ☐ Interactive questions with clinicians or researchers
- ☐ Large-print version with less text
- ☐ Captions, audio, or other accessible versions
- ☐ Other: ____

公众认知调查

新型生物医学技术与阿尔茨海默病治疗



Q01 您的年龄是?

单选题：选1项

- ○ 18岁以下
- ○ 18–30岁
- ○ 31–45岁
- ○ 46–60岁
- ○ 61–70岁
- ○ 71–80岁
- ○ 81岁及以上

Q02 您是否接受过生命科学、医学、药学或其他相关专业教育?

单选题：选1项

- ○ 是，目前正在学习或从事相关领域
- ○ 曾经接受过相关专业教育，但目前不从事相关领域
- ○ 没有相关专业背景
- ○ 不确定

Q03 您本人或身边亲近的人是否有过阿尔茨海默病或其他认知障碍相关经历?

单选题：选1项

- ○ 有较直接的经历，例如本人、家人患病或参与照护
- ○ 有一定接触或了解，但没有长期照护经历
- ○ 没有相关经历
- ○ 不愿回答

Q04 在填写本问卷之前，您对“RNA编辑”的了解程度最接近以下哪一种?

单选题：选1项

- ○ 从未听说过
- ○ 听说过这个词，但不知道是什么意思
- ○ 知道大致用途或基本概念
- ○ 能够简单解释其基本原理及可能用途
- ○ 对相关技术比较熟悉

Q05 对于下面几个说法，请根据您现在的理解进行判断。

每个说法选1项

1. RNA编辑可以在不直接改写基因组DNA的情况下，改变特定RNA中的信息。

○ 比较符合我的理解 ○ 比较不符合我的理解 ○ 不确定

2. 不同RNA编辑方法在作用持续时间、安全性和技术成熟度等方面可能存在差异。

○ 比较符合我的理解 ○ 比较不符合我的理解 ○ 不确定

3. 判断一种RNA编辑疗法是否适合患者，仅了解它的作用原理通常是不够的。

○ 比较符合我的理解 ○ 比较不符合我的理解 ○ 不确定

4. 如果一种RNA编辑技术在动物实验中改善了相关指标，通常就可以认为它对患者有效。

○ 比较符合我的理解 ○ 比较不符合我的理解 ○ 不确定

Q06 假设一项针对阿尔茨海默病的新型RNA编辑治疗研究报告：

多选题

治疗后，与疾病相关的一项生物标志物平均下降了40%。

仅根据这项结果，您认为下面哪种理解最合适?

- [ ] 说明该治疗已经产生了较明确的临床治疗效果
- [ ] 说明该治疗可能产生了预期的生物学作用，但还需要观察患者症状和长期结局
- [ ] 说明该治疗值得使用，但是否适合具体患者主要取决于医生判断
- [ ] 目前的信息主要反映了一个研究指标的变化，还不足以判断其临床价值
- [ ] 不确定

Q07 随后研究人员又公布了以下信息：

单选题：选1项

这项治疗已经进行了初步人体研究。部分患者的相关生物学指标得到改善，但研究人数仍较少，对于认知功能改善程度以及长期不良反应，目前还需要进一步随访。

如果您看到这样的研究结果，您对这项技术现阶段的判断更接近：

- ○ 已经有一定依据认为它具有研究价值，但是否具有足够临床获益仍需要进一步验证
- ○ 既然已经在人体上观察到生物学变化，可以初步认为治疗价值已经得到证明
- ○ 目前最重要的是确认它能否进一步改善与疾病有关的生物学指标
- ○ 是否值得进一步使用，主要取决于患者个人对新技术的接受程度
- ○ 我还需要更多信息才能形成判断

Q08 如果您或家人未来需要考虑一种尚不熟悉的新型阿尔茨海默病治疗，在做决定前，您认为下面哪些信息最值得优先确认？

多选题：最多选4项

- [ ] 相比目前已有治疗，它实际能够改善或延缓多少病情
- [ ] 严重不良反应的类型以及发生概率
- [ ] 相关结论来自多大规模、什么阶段的人体研究
- [ ] 哪些患者可能获益，哪些患者可能并不适用
- [ ] 治疗作用的分子机制是否足够明确
- [ ] 治疗效果可以维持多久
- [ ] 如果不接受这种治疗，还有哪些替代方案
- [ ] 治疗机构或研发团队的专业声誉
- [ ] 已经接受治疗的患者个人体验
- [ ] 治疗费用以及长期经济负担

Q09 对一种新的治疗方案，下面哪种信息表达方式最有助于您理解它的风险？

单选题：选1项

- ○ “目前研究显示整体安全性较好，但仍存在一定风险”
- ○ “研究中约每100名接受治疗者中有3人出现某项严重不良反应，同时说明研究总人数和随访时间”
- ○ “严重不良反应发生率较低，大部分受试者都能耐受”
- ○ “目前没有发现超出研究人员预期的安全性问题”
- ○ 以上信息需要结合起来，我才比较容易判断
- ○ 不确定

Q10 当您了解一种不熟悉的生物医学技术时，哪些情况最容易妨碍您理解或作出判断？

多选题：最多选3项

- [ ] 专业术语太多
- [ ] 没有说明研究处于哪个阶段
- [ ] 只说“有效”，没有说明具体改善了什么
- [ ] 只说风险“较低”，没有给出具体人数或概率
- [ ] 信息过长、重点不清楚
- [ ] 图表或数据难以理解
- [ ] 不知道信息来源是否可靠
- [ ] 缺少与现有治疗及“不治疗”的比较
- [ ] 缺少字幕、语音、图示或易读版本
- [ ] 没有可信的人可以进一步询问
- [ ] 其他：____
- [ ] 暂时没有明显障碍

Q11 如果我们制作相关科普或知情同意材料，哪些形式会更方便您理解？

多选题

- [ ] 一页纸简明版
- [ ] 图示/流程图
- [ ] 短视频
- [ ] 专家详细解释
- [ ] 真实数字和绝对风险示例
- [ ] 常见问题清单
- [ ] 可与医生或研究人员互动提问
- [ ] 大字号、低密度文字版本
- [ ] 字幕、音频或其他无障碍版本
- [ ] 其他：____